Introduction to DATASCI 406

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Computational Methods in Statistics and Data Science (DATASCI 406)

Course Overview

Course Goals

At the end of this course you will be able to

- Trade **computation time** for **analytical effort**
- Connect **mathematical theory** to **computational applications**
- Understand and explain **costs, benefits, and limits** of computational techniques
- Implement key **statistical algorithms** in R
- Create your own **statistical models**
- **Analyze data** using computational techniques

What is Computational Statistics?

Statistics: the application of **probability theory** to real **data**.

Goals:

- Specifying **models of data generation**
- Performing **inference** to connect real data to model
  - **Estimate** model parameters
  - **Test** hypotheses
  - **Predict** future observations
- Quantify **uncertainty** in the inference process.
- Understand **operating characteristics** of tools.

Computational statistics uses **substantial amounts of computation** to achieve these goals.

Two fundamental concepts in computational statistics

Monte Carlo: replace a quantity you cannot compute with an average over random draws. If we can get a sample $X_1, \ldots, X_n$, then

$$\mathrm{E}\left(g(X)\right) = \int g(x) f(x)\, dx \;\approx\; \frac{1}{n} \sum_{i=1}^{n} g(X_i)$$

Optimization: search for the parameter value that makes something best, when you cannot solve for it algebraically.

E.g., for a complicated density f with parameter θ, what value of θ would be most consistent with my sample?

Four Modules

Module	Name	Topic
1	**Integrate**	Monte Carlo with the distributions you have
2	**Generate**	Monte Carlo when you need a new distribution
3	**Resample**	Monte Carlo with only your data
4	**Fit**	Optimization

Let's see some examples!

Module 1: Integrate

A density

Consider a **random variable** X with density f:

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{1.5\,x - 0.25\,x^2 - 2.5966\right\}$$

How would you find $\mathrm{E}(X) = \int_{-\infty}^{\infty} x f(x)\,dx$?

Nothing here looks like an integral you can do by hand.

Draw, then average

Suppose I hand you `sampleX(n)`, which returns n draws from f.

```
x <- sampleX(100000)
mean(x)
```

```
## [1] 2.999
```

More complicated integrals

$$\mathrm{E}\left(|X|\right) = \int_{-\infty}^{\infty} |x|\, f(x)\, dx$$

$$\Pr(X > 5) = \int_{5}^{\infty} f(x)\, dx$$

$$\mathrm{E}\left(\log(1 + e^{X})\right) = \int_{-\infty}^{\infty} \log(1 + e^{x})\, f(x)\, dx$$

$$\mathrm{E}\left((1 + X^2)^{-1}\right) = \int_{-\infty}^{\infty} \frac{f(x)}{1 + x^2}\, dx$$

Same code, different g

Each is $\mathrm{E}(g(X)) = \int g(x)f(x)\,dx$ for a different g.

```
c(EabsX  = mean(abs(x)),
  P_gt_5 = mean(x > 5),
  Esoft  = mean(log1p(exp(x))),
  Einv   = mean(1 / (1 + x^2)))
```

```
##    EabsX  P_gt_5   Esoft    Einv
## 3.01620 0.07904 3.10634 0.17436
```

Change g, keep everything else.

Sampling in R

Every common distribution has an `r` function:

`rnorm`	Normal	`rbinom`	Binomial
`runif`	Uniform	`rpois`	Poisson
`rexp`	Exponential	`rgamma`	Gamma
`rt`	Student's t	`rbeta`	Beta

All take the number of draws first: `rnorm(n, mean, sd)`.

```
sampleX
```

```
## function (n)
## rnorm(n, mean = 3, sd = sqrt(2))
```

Functions of samples

A **statistic** is a function of a sample:

$$T = T(X_1, X_2, \ldots, X_n)$$

We use statistics to

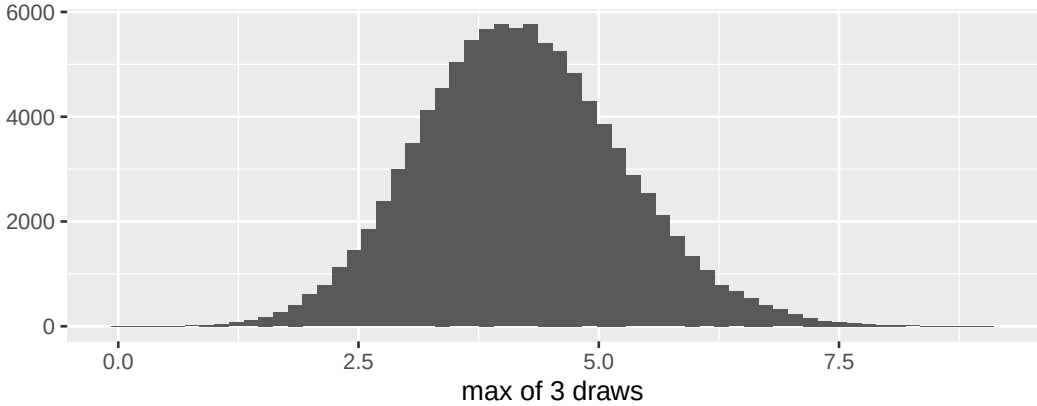
- Estimate population parameters
- Test hypotheses about populations
- Perform prediction for new observations

A statistic is a random variable, so it has a distribution.

Distributions of statistics

$$T = \max_{i=1,2,3} X_i$$

```
sample_max <- function(n) max(sampleX(n))
empirical_distrib <- replicate(100000, sample_max(3))
```


6000
4000
2000
0
0.0
2.5
5.0
7.5
max of 3 draws

Some Properties

```
mean(empirical_distrib); var(empirical_distrib)
```

```
## [1] 4.195
```

```
## [1] 1.124
```

```
quantile(empirical_distrib, c(0.25, 0.5, 0.75))
```

```
##   25%   50%   75%
## 3.465 4.157 4.882
```

These are **estimates**. How far off? That is module 1.

Module 2: Generate

A distribution R does not have

The **Pareto** distribution: incomes, city sizes, file sizes. A heavy right tail, so extreme values are far more common than under a Normal.

For shape $\alpha > 0$ and $x \geq 1$:

$$\Pr(X > x) = x^{-\alpha}$$

Therefore it has a **cumulative distributiuon function** (CDF)

$$F(x) = 1 - x^{-\alpha}, \qquad x \geq 1$$

The quantile function

Given a probability u, which x has $F(x) = u$?

$$u = 1 - x^{-\alpha} \quad \Longrightarrow \quad x^{-\alpha} = 1 - u \quad \Longrightarrow \quad F^{-1}(u) = (1-u)^{-1/\alpha}$$

$F^{-1}(0.5)$ is the median, $F^{-1}(0.9)$ the 90th percentile.

```
qpareto <- function(u, alpha = 3) (1 - u)^(-1/alpha)
qpareto(c(0.5, 0.9, 0.99))
```

```
## [1] 1.260 2.154 4.642
```

How to sample? Inversion!

Later we will prove: If $U \sim \text{Unif}(0, 1)$, then $F^{-1}(U)$ has CDF F.

Algorithm in R:

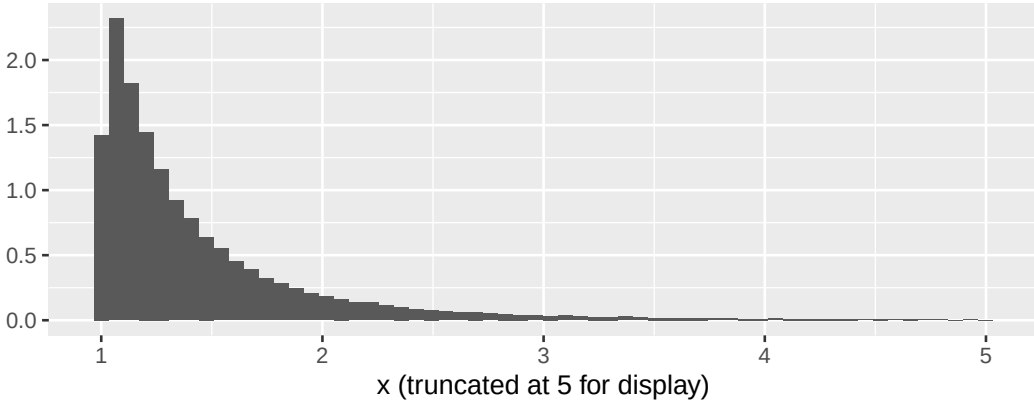
- Sample U using `runif`
- Create X using $F^{-1}(U)$

A Pareto sampler in three lines

```
rpareto <- function(n, alpha = 3) qpareto(runif(n), alpha)

x <- rpareto(100000)
c(mean = mean(x), exact = 3 / 2, min = min(x))
```

```
##  mean exact   min
## 1.502 1.500 1.000
```


2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
x (truncated at 5 for display)

Module 3: Resample

Unknown distributions

In the previous examples, we had **known distributions** (later we will see these in hypothesis tests and estimating parameters of assumed population distributions).

What about situations where we **don't want to assume a population distribution?**

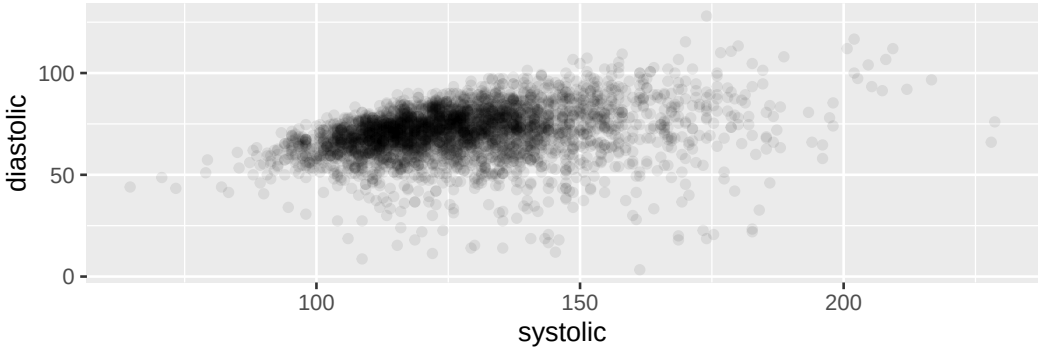
Idea: Instead of writing down a CDF F, what if we estimated it?

What distribution is this?

Systolic and diastolic blood pressure, NHANES 2013–14.

```
load("../../data/aspirin.rda")
dim(combined)
```

```
## [1] 3554    5
```



No named family fits

```
##  systolic diastolic
##    0.8871   -0.6222
```

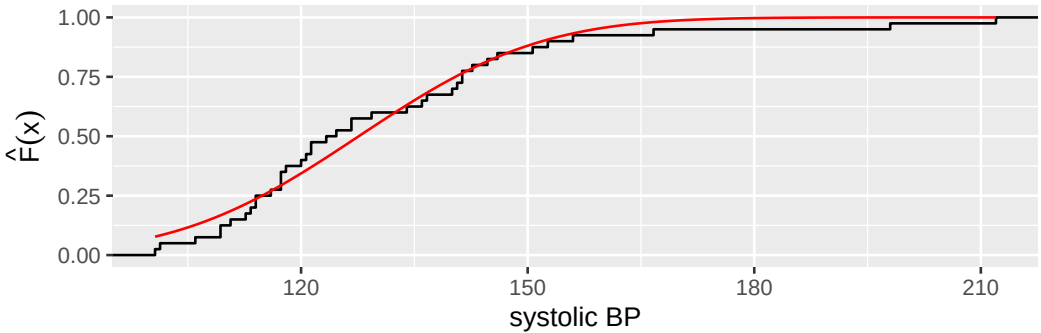
The two margins skew in **opposite directions**, so this is not bivariate Normal. There is also a clump of implausibly low diastolic readings.

We could go hunting for a family that fits. Or we could stop assuming one.

Estimating the CDF

The **empirical CDF** puts mass $1/n$ on each observed value:

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{X_i \leq x\}$$



Inversion, one module later: The Bootstrap

Module 2: to sample from F, apply F^{-1} to a uniform draw.

Now do exactly that to $\hat{F}$. Since $\hat{F}$ puts mass $1/n$ on each observed point, $\hat{F}^{-1}(U)$ returns **one of your data points, chosen uniformly at random.**

$$\text{sampling from } \hat{F} \;=\; \texttt{sample(x, replace = TRUE)}$$

We call this "bootstrapping.'' It is inversion applied to the CDF you estimated.

Correlation of systolic and diastolic BP

Blood pressure, NHANES 2013–14:

```
theta <- cor(combined$sys_mean, combined$dia_mean)
c(n = nrow(combined), r = theta)
```

```
##         n         r
## 3554.000     0.293
```

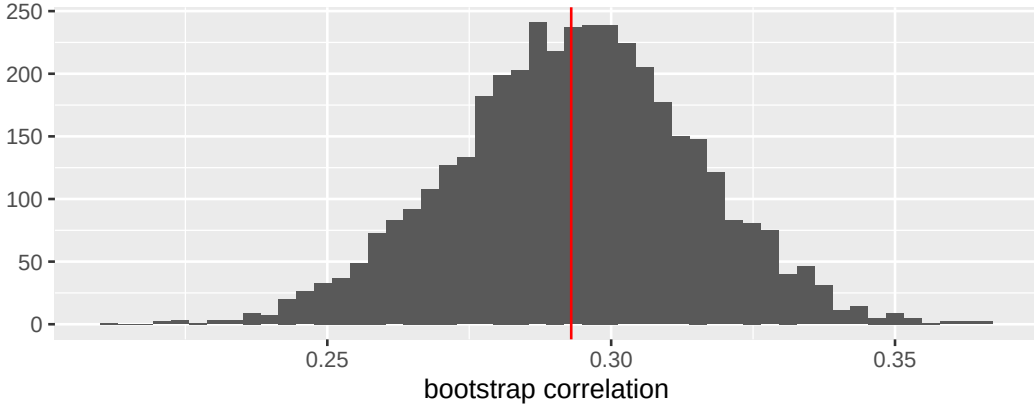
What is the **standard error of that correlation**? What would be a good **95% confidence interval?**

Bootstrapping for SE, quantiles

Treat the sample as the population and draw from it.

```
set.seed(406)
boot_r <- replicate(4000, {
    i <- sample(nrow(combined), replace = TRUE)    # rows, not values
    cor(combined$sys_mean[i], combined$dia_mean[i])
})
c(se = sd(boot_r), quantile(boot_r, c(0.025, 0.975)))
```

```
##       se    2.5%   97.5%
## 0.02137 0.25020 0.33478
```


250
200
150
100
50
0
0.25
0.30
0.35
bootstrap correlation

Module 4: Fit

Only the maximum survives

Same $X \sim f$ as before, $\sigma^2 = 2$ known. We draw $m = 200$ groups of three, but the record keeps only the largest value in each group:

$$M_j = \max\left(X_{j1},\, X_{j2},\, X_{j3}\right), \qquad j = 1, \ldots, 200$$

```
set.seed(406)
m <- 200
groups <- matrix(sampleX(m * 3), nrow = m)
M <- apply(groups, 1, max)   # all we get to see
```

Estimate μ.

How to estimate?

Note that μ is not the same as $\mathrm{E}(\max X)$:

```
mean(M)
```

```
## [1] 4.299
```

1. Maximum likelihood estimation
2. Algorithm that converges to the MLE

Maximum likelihood, briefly

The **likelihood** is the probability of the data we saw, read as a function of the unknown parameter:

$$L(\mu) = \prod_{j=1}^{m} f_M(M_j; \mu) \qquad \hat{\mu} = \arg\max_{\mu} \; L(\mu)$$

Pick the μ that makes what we observed as likely as possible. More on this in the statistical review.

Why not just maximize the likelihood?

The max of three draws has density $3f(x)F(x)^2$, so

$$\ell(\mu) = \sum_{j=1}^{m} \left[\log \phi\Big(\tfrac{M_j - \mu}{\sigma}\Big) + 2 \log \Phi\Big(\tfrac{M_j - \mu}{\sigma}\Big) \right]$$

Setting $\ell'(\mu) = 0$ gives

$$\sum_{j=1}^{m} \frac{M_j - \mu}{\sigma^2} \;=\; \frac{2}{\sigma} \sum_{j=1}^{m} \frac{\phi\Big(\tfrac{M_j - \mu}{\sigma}\Big)}{\Phi\Big(\tfrac{M_j - \mu}{\sigma}\Big)}$$

μ is stuck inside Φ, which is itself an integral with no closed form. **You cannot solve this for μ.**

What we would do with the full data

With all $3m$ values in hand, estimating μ is a sample mean:

$$\hat{\mu} = \frac{1}{3m} \sum_{j=1}^{m} \underbrace{\sum_{i=1}^{3} X_{ji}}_{S_j}$$

We only need the **group totals** S_j, not the individual values.

We cannot compute S_j. But we know one of its three terms and have a bound on the other two.

Making a reasonable guess

Replace the total we cannot compute with **its expectation given what we saw.**

Given M_j, one value equals M_j and the other two are draws from f conditioned to lie below M_j:

$$\mathrm{E}\,(S_j \mid M_j) \;=\; M_j \;+\; 2\,\mathrm{E}\,(X \mid X \le M_j)$$

For a Normal the conditional mean is closed form:

$$\mathrm{E}\,(X \mid X \le a) \;=\; \mu - \sigma\,\frac{\phi\left(\frac{a-\mu}{\sigma}\right)}{\Phi\left(\frac{a-\mu}{\sigma}\right)} \;<\; \mu$$

It depends on μ, which is what we are trying to find. So guess, then iterate.

Example

Start from the observed average maximum, $\mu^{(0)} = 4.30$, and take the first group, which came back with $M_j = 5.77$:

$$\mathrm{E}\left(X \mid X \leq 5.77\right) = 3.91$$

$$\mathrm{E}\left(S_j \mid M_j\right) = 5.77 + 2(3.91) = 13.60$$

The true total was $1.16 + 2.57 + 5.77 = 9.50$.

One group is badly wrong. The M step averages 200 of them.

EM: Expectation-Maximization algorithm

Start with a guess $\mu^{(0)}$. For $t = 1, 2, \ldots$:

1. **E**: compute $\mathrm{E}(S_j \mid M_j)$ for every group, using $\mu^{(t-1)}$.
2. **M**: set $\mu^{(t)} = \frac{1}{3m} \sum_j \mathrm{E}(S_j \mid M_j)$.
3. Repeat until it **convergence** (estimate stops moving).

Starting from $\mu^{(0)} = 4.30$, one step gives 3.49.

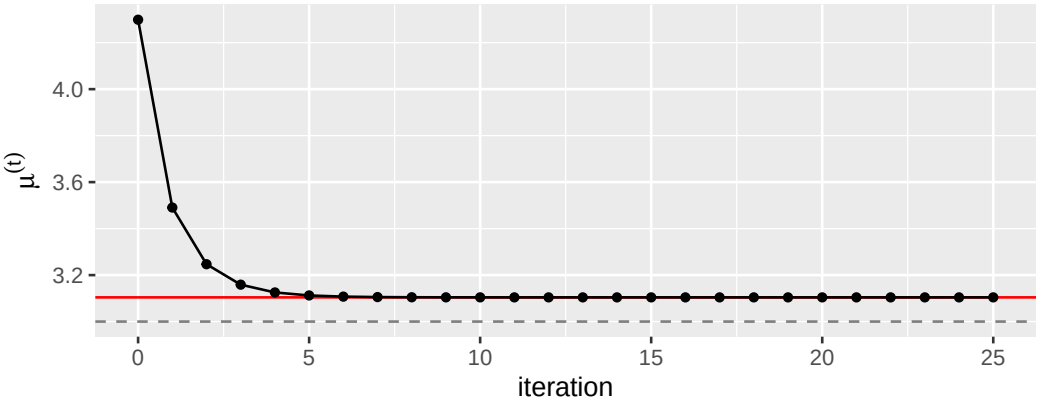
EM in R

```
sigma <- sqrt(2)
Etrunc <- function(a, mu, s) mu - s * dnorm((a-mu)/s) / pnorm((a-mu)/s)

guess <- mean(M)
for (i in 1:100) {
    e_step <- (M + 2 * Etrunc(M, guess, sigma)) / 3
    m_step <- mean(e_step)
    guess <- m_step
}
c(EM = m_step, naive = mean(M), mu = 3)
```

```
##    EM naive    mu
## 3.104 4.299 3.000
```

Convergence



Red is the **MLE from the data we actually have**; dashed grey is $\mu = 3$.

What is EM converging to?

Not μ, and not the average of the values we lost. EM maximizes the likelihood of the data **we observed**:

population μ	3.000
mean of the 600 hidden values	3.063
MLE from the 200 maxima	**3.104**
EM after 25 steps	3.104

EM lands on the MLE, exactly. The gap from 3 is **sampling error in 200 maxima**, not a flaw in the algorithm. With more groups the MLE moves toward 3.

Course Logistics

People

Professor	Mark Fredrickson (<mfredric@umich.edu>)
GSIs	Ishan Paul (<ishanpl@umich.edu>), Qingyuan Yu (<qyyu@umich.edu>)
Student Hours	TBA, see Canvas

Prerequisites

- Some programming experience (R, Python, C/C++, Java)
- An understanding of the following terms: Distributions and random variables, random sampling and sampling distributions, population parameters and estimation, hypothesis tests, basic calculus (integrals, derivatives).

Course Materials

No required text, but several **recommended** (particularly first 2). Links to electronic copies available through the library on Canvas.

- Rizzo, M. L. *Statistical Computing with R*, 2nd ed.
- Wickham, H. & Grolemund, G. *R for Data Science*
- Robert, C. P. & Casella, G. *Introducing Monte Carlo Methods with R*
- Irizarry, R. A. *Introduction to Data Science*
- Agresti, A. *Foundations of Linear and Generalized Linear Models*
- Gentle, J. E. *Computational Statistics*
- Härdle, W., Gentle, J. E. & Mori, Y., eds. *Handbook of Computational Statistics*

All slides and videos will be posted to **Canvas** and **OneNote**.

Approach

The class is composed of **four modules** and uses a **flipped classroom** approach.

Each module consists of short **check-in quizzes**, a **project**, and an **exam**. There is no separate final exam.

Materials are presented in **video lectures**. Class time is for short **check-in quizzes**, working on examples and the project, and exams.

Quizzes

- 4–6 multiple choice plus one short answer, every session
- Covers the material you watched **before** class
- **Lowest per module dropped. No makeups.**
- I will reach out to students with testing accommodations.

Projects

Each module includes a project. The project gives you an opportunity to:

- Implement key statistical algorithms
- Investigate the properties of those algorithms
- Apply these techniques to real-world data

These projects become your notes for your exams. **You will be asked questions on the exams about your project.**

Exams

Each module concludes with an exam. Approximate coverage:

- 50%: General conceptual questions, similar to the quizzes.
- 50%: Questions about the results of your own project.

You may bring any **paper** notes you wish. No electronic devices. Held during class periods, with the exception of the Module 4 exam, which uses the scheduled final exam time.

Students with accommodations will be contacted with alternate exam location information.

Grading

Each module counts for 25% of your grade. Within each module:

- Exam: 20% of overall grade
- Quizzes: 5% of overrall grade, one drop per module

AI policy

Permitted and expected *on projects and all out-of-class work.*

Not permitted *on quizzes or exams.*

Remember: you will be asked about project details. Be prepared to explain coding details, even if you used AI to assist.

Other Policies

- Course Recordings
- Disability Statement
- Mental Health and Well-Being
- Sexual Misconduct Policy

Tour of Canvas

Next Session

- No quiz
- Video: statistical review
- Class period: Introduction to R and RMarkdown
- All students: install R and RStudio on laptops, directions on Canvas